

Fundamentals of Financial Markets

Week #2: The Money Market

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Focus:

This week's main focus is on instruments and contracts used for short-term (0 - 1 year, mostly 1 - 90 days) funding/lending purposes. The financial institutions' borrowing and lending activities are everyday problems in the fiat currency world: modern money is created as debt, thus the most important feature is the interest.

Concepts:

- interbank deposit market: loans and deposits are the two sides of the same deal
- borrowing = long deposit position
- lending = short deposit position
- Federal Funds = Fed funds: commercial banks trade their reserve balance at the Fed, similar to the interbank deposit market, but only banks with account at the Fed can trade
- Federal Funds Rate = the overnight rate of the Fed Funds
- the bank discount method convention (ACT/360 for USD, EUR and HUF; ACT/365 for GBP)
- linear interests are linear in 3 ways: time, notional and rate
- two-sided market prices
- LIBOR (London Interbank Offered Rate), EURIBOR
- Eurodollar deposits
- Certificate of Deposits: a securitized time deposit issued by a bank; better than a normal time deposit because it is a security and thus it is tradeable during its lifetime
- Commercial Paper: a short term note (security) issued by any company (mostly by nonfinancials); tradeable during its lifetime, could be secured by a bank's credit line
- repo (sale-and repurchase agreement)
- active repo / passive (reverse) repo
- haircut
- central banks use repos as a monetary policy tools¹
- derivative and cash products
- Treasury - to - EuroDollar Rate = TED-spread = 3Months LIBOR – 3Months T-Bill
- some investments give some additional benefits over the pure cash flow of the structure: a liquid secondary market (if the instrument is tradeable), lower counterparty risk, being able to repo...etc. The markets perception about the value of these benefits depends on the actual market sentiment: in a tense, volatile market these features bear more value, so TED-spread will increase.

INSIGHT: *Later during the financial markets related courses you will see a lots of money market related derivatives. Why are there so many derivatives? Mainly because deposit deals are very*

¹ The New York Fed conducts repo and reverse repo operations each day as a means to help keep the federal funds rate in the target range set by the Federal Open Market Committee (FOMC). When there is a Fed rate decision it is about the upper end of this target range.

inconvenient way most of the time (counterparty risk, limits, non-cancellable / non-tradeable feature...etc). There is a strong motivation to avoid trading deposits whenever you can substitute them by other products. Even if you have no better solution and thus you have to trade deposit, you try to trade short dated ones.

Problems:

4.1.) A dealer quotes the following yield for a 85 days maturing T-Bill: 3.055 / 2.950.

4.1. a.) How much dollars would you get if you could sell the 85-days T-Bills in USD 1 million notional?

4.1. b.) How much should a buyer pay when she purchases the 85-days T-Bills in USD 1 million notional?

4.1. c.) What would be the effective yield on your investment if you would buy a 85-days T-Bill today and would keep it until maturity?

4.1. d.) Assume you have 1 million notional of the 85-days T-Bill. What is the highest quoted bid rate on which you can sell the T-Bills if you would like to get at least USD 990.000,-?

4.1. e.) Assume you purchase 1 million notional of the 85-days T-Bill today. What is the highest bid rate on which you can sell the T-Bills 30 days later and get back your initial invested dollar amount?

4.2.) On the last Friday (30th September 2022) the following quotes were seen:

USD deposit O/N (overnight)	3.05/3.12
USD deposit 1W (one week)	3.20/3.25
USD deposit 3M (three months)	3.70/3.75
3-Months T-Bill	3.25/3.20

4.2.a.) If a bank would like to borrow 10 million USD O/N, what would be the cash flow of the deal?

4.2.b.) If a bank would like to lend 10 million USD for 1 week, what would be the cash flow of the deal?

4.2.c.) How much is the TED-spread now? (estimation)

4.2.d.) A bank holds 3 Months T-Bills in 10 million notional and decides to sell it and lend the money on the deposit market. At the end of the 3-months period, how much extra dollars would be generated by this transaction compared to the original buy-and-hold the T-Bill situation?

4.2.e.) If investors can make more money by selling their T-Bills and lending the proceeds at the money markets, why anyone still holds T-Bills?

4.2.f.) What would happen to the TED-spread if more and more investors would sell the T-Bills and lend their money on the deposit markets?

4.2.g.) What happened to the TED-spread historically during crises times?

4.3. What would you expect to happen to the spread between yields on commercial paper and Treasury bills if the economy were to enter a steep recession? (Source: Bodie-Kane_Marcus: Investments, Chapter 2 problem set)

4.4. Which of the following correctly describes a repurchase agreement? (Source: Bodie-Kane_Marcus: Investments, Chapter 2 problem set)

- a. The sale of a security with a commitment to repurchase the same security at a specified future date and a designated price.
- b. The sale of a security with a commitment to repurchase the same security at a future date left unspecified, at a designated price.
- c. The purchase of a security with a commitment to purchase more of the same security at a specified future date.

4.5.) A fund manager holds a 3-years-long Treasury Note (T-Note²) in 2 million USD notional. The market for this T-Note has a net price quote of 93.87% / 93.90%. The T-Note has an accrued interest of 0.19%. The T-Note pays coupon semi-annually, the next coupon payment will happen on 15th February 2023. The 1-month LIBOR is at 3,14%. The fund manager needs extra 1 million USD liquidity for 1 months.

4.5.a.) How much money could the fund manager raise by selling all the T-Notes?

4.5.b.) In how much notional should she sell the T-Notes to raise at least 1 million USD?

4.5.c.) She could borrow 1 million USD at LIBOR + 150 bp from her bank. How much interest would she pay at the end of the 1-months period?

4.5.d.) The same bank offers her a repo deal based on her T-Notes. The bank requests a 10% haircut from the T-Note price and offers a LIBOR + 25 bp repo rate for 1 months. In how much notional should she use her T-Notes to get the 1 million USD funding (please round up to ten thousands)? What would be the cash flow of the fund manager if she enters into this deal?

4.5.e.) Which counterparty is in active repo position and which one is in passive (reverse) repo position?

4.5.e.) The bank requested a haircut. Why?

4.6.) A Treasury Bond maturing on the 15th February 2052 (CUSIP number: 912810TD0) pays coupon on each 15th February and 15th August until it matures. The nominal interest is 2 1/4% (= two-and-a-quarter percent = 2,25% annually, so on each coupon date it pays only 1,125% nominal interest semi-annually).

On the 30th September 2022 the following market quotes were seen:

² This is a real life existing T-Note, you can check all the features on Bloomberg to learn more, CUSIP: 91282CDZ1. All the prices were real prices on 30th September 2022.

T-Bond 2052 (912810TD0) net price 73.3437 / 73.3750
6-Months LIBOR 4.2320

4.6.a.) What is the gross price of the T-Bond? (Last time it paid coupon on the 15th August, 46 days ago)

4.6.b.) How much interest will be paid on the 15th February 2023 to an investor who holds 1 million USD notional of this T-Bond?

4.6.c.) An investor who holds 1 million T-Bond would like to enter into a 6-months repo deal with her bank to get liquidity. The bank offers LIBOR + 50 bp repo rate and asks for a 20% haircut. What would be the cash flow of the investor if she enters into this deal?

4.6.d.) The investor could have sold the T-Bonds on the market, collect the liquidity and 6 months later she could buy back the T-Bonds? Why could a repo be better than selling the T-Bonds now at the market and buying them back later on the market?